

Data Analysis with Python

Matrix Algebra, Vectors and Vector Spaces

Group assignment: Max. of 5 members per group

Deadline: October 20, 2026, 11:59 PM EAT

Purpose. Investigate synthetic mobile money transactions using pandas, NumPy, and plots.

Resources. Download the CSV from [Synthetic Mobile Money Transaction Dataset, Mendeley Data \(repository version 2\)](#); DOI: [10.17632/zhj366m53p.2](#). Read D. Azamuke, M. Katarahweire and E. Bainomugisha, *A labeled synthetic mobile money transaction dataset*, Data in Brief 60 (2025), 111534: [research paper](#); DOI: [10.1016/j.dib.2025.111534](#).

Data orientation. MoMTSim generates simulated transactions that mimic the statistical qualities of source data. Example columns in the dataset include: `step`, `transactionType`, `amount`, `initiator`, `oldBalInitiator`, `newBalInitiator`, `recipient`, `oldBalRecipient`, `newBalRecipient`, and `isFraud`. Treat account IDs as strings and the fraud label as an outcome, not a feature. Consult the research paper for field meanings.

1. Load, audit, and explore the data.

- Describe the simulation and the meaning of the fraud label.
- Report dimensions, data types, missing values, exact duplicate rows, and invalid labels. Document your treatment of each issue; investigate negative balances before deciding whether to remove them.
- Produce a transaction-type summary with counts, total amounts, median amounts, and fraud rates. Include a labeled transaction-type count plot and an amount-distribution plot. Discuss skew and class imbalance; distinguish transaction counts from total monetary value.

Starter code

```
import pandas as pd
import numpy as np
df = pd.read_csv("synthetic_mobile_money_transaction_dataset.csv",
                 dtype={"initiator": "string", "recipient": "string"})
print(df.shape)
print(df.isna().sum())
summary = df.groupby("transactionType").agg(
    count=("amount", "size"), total=("amount", "sum"),
    median=("amount", "median"), fraud_rate=("isFraud", "mean"))
analysis_df = df # revise after your audit; document any cleaning
```

2. Represent transactions as matrices and vectors.

Take a random sample of up to 2,000 rows from the cleaned data. For transaction i , define the column vector and the matrix with transactions as rows:

$$x_i = (a_i, b_{I,i}^{\text{old}}, b_{I,i}^{\text{new}}, b_{R,i}^{\text{old}}, b_{R,i}^{\text{new}})^T \in \mathbb{R}^5, \quad X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n \times 5}.$$

- Display three actual rows of X and label their coordinates. State the dimensions and meanings of X , X^T , $X^T X$ and $X X^T$. Explain why account IDs and transaction types are excluded from this numeric representation.
- Compute the mean vector $\mu = X^T \mathbf{1}/n$. Verify it against pandas means. Express the total amount as a dot product with $\mathbf{1}$; explain why a column of X belongs to $\mathbb{R}^n$, whereas x_i belongs to $\mathbb{R}^5$.
- Choose two nonzero standardized transaction vectors u, v . Compute $\|u\|_2$, $\|u - v\|_2$, $u^T v$, cosine similarity and their angle. Interpret these and compare raw and standardized distances. Explain why similarity alone does not determine fraud.

```
sample = analysis_df.sample(n=min(2000, len(analysis_df)),
                           random_state=1105).copy()
cols = ["amount", "oldBalInitiator", "newBalInitiator",
        "oldBalRecipient", "newBalRecipient"]
X = sample[cols].to_numpy(dtype=float)
mu = X.mean(axis=0)
s = X.std(axis=0, ddof=1)
# TODO: standardize X to define Z before continuing
u, v = Z[0], Z[1] # check both norms before division
distance = np.linalg.norm(u - v)
```

3. Scenario: operations monitoring and transfer investigation.

You are an analyst for a mobile money provider. The operations manager wants to identify busy periods, plan support capacity, and investigate unusual transfers. Use the provided transaction records to make data-driven recommendations.

- Build a step-by-type count matrix $M \in \mathbb{R}^{S \times 5}$ from the full cleaned data. Label its rows and columns; use $M\mathbf{1}_5$ to obtain the step-count vector. Plot counts against step and identify the three busiest steps and their dominant transaction types.
- For sampled TRANSFER records, calculate balance changes using $Y = X_T B^T$, where X_T contains only transfer rows of X . Verify the result by direct subtraction and inspect the residuals $r_I = \Delta b_I + a$ and $r_R = \Delta b_R - a$, using 0.01 monetary units as a declared tolerance. Compare mismatch proportions for fraud and legitimate labels, reporting group counts.
- From the full cleaned data, calculate each step's fraud count and rate. Plot rates with step counts available for context. Compare a busy step with a high-fraud-rate step and explain why high volume and high risk need different responses.
- Recommend two actions for the manager, based on your findings from the numerical computations.

Balance-change guide: $B = \begin{bmatrix} 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}$, $Bx_i = (\Delta b_{I,i}, \Delta b_{R,i})^T$.

4. Scenario: predict next-step transaction volume.

The provider wants a simple forecast to guide staffing and processing capacity. Let c_t be the transaction count at step t in the full cleaned data. Fit $\hat{c}_{t+1} = \beta_0 + \beta_1 c_t$, using design rows $(1, c_t)$ and targets c_{t+1} . Work in simulation steps (1 step = 1 hour).

- Check that steps are consecutive; investigate any gaps before treating them as zero-count periods. Sort steps and create adjacent input–target pairs. Use the earliest 70% of pairs for training and the remaining 30% for testing. State the split boundary, matrix dimensions, and why randomly splitting these pairs would be inappropriate for future prediction.
- Fit $A_{\text{train}}\beta \approx y_{\text{train}}$ with least squares and explain $\hat{y} = A_{\text{test}}\beta$ as a vector in the span of the columns of A_{test} . Report coefficients. Evaluate held-out mean absolute error (MAE) and root mean squared error (RMSE); compare with the persistence baseline $\hat{c}_{t+1} = c_t$ on the same test pairs.
- Plot observed and predicted test counts against target step; also show the baseline. Explain whether the model improves on it and how errors affect staffing. Discuss your findings.

Submission and presentation

- Write a coherent report answering Questions 1–4.
- Record a 5–10-minute group video presenting the solution. Every member should speak and explain a computation or interpretation. Upload to **YouTube (unlisted is acceptable)** or **Google Drive**; include a clickable viewing link on the report's first page. Ensure the assessor has viewing access.